APPIAN INTERVIEW EXPERIENCE

Appian is a software company that automates business processes. The Appian AI-Powered Process Platform includes everything you need to design, automate, and optimize even the most complex processes, from start to finish. The world's most innovative organizations trust Appian to improve their workflows, unify data, and optimize operations, resulting in better growth and superior customer experiences. For more information, visit appian.com.

Roles Offered: Software Engineer, Quality Engineer, UI/UX Designer, Product Manager
Stipend: ₹1,50,000 INR (all roles)
Selection Process:
Pre-Placement Talk (PPT) → Online Assessment → Technical Interview 1 → Technical Interview 2 → HR Interview

Roles Applied For: Quality Engineer, Software Engineer, and UI/UX Designer
Final Offer: Selected for the **Quality Engineer** role

Online Assessment (for the role of QE):

Test platform: Glider AI

It consisted of 25 MCQs focused on:

• Aptitude and logical reasoning

• Programming fundamentals

The test lasted for 45 mins and I had the option to answer the questions in any order.
Most of them involved quick thinking and focused on paying attention to detail.

These are some of the question categories:

Aptitude – puzzles, finding duration / time, relation between family members
Finding output/error in the given code snippet
Expression evaluation and operators
Reference variables and pointers

Based on this online assessment, 17 people were shortlisted for the next round.

Technical Interview – 1: (30-45 mins)

The interviews were conducted at CUIC. The interviewer started off by going through my resume and appreciated me about the various activities that I'd done in two years (I was a part of clubs, completed some certifications (NPTEL, Hackerrank) and a JAVA training program. I had also done a summer internship at DCSE, CEG). I was asked to talk about any of these.

I spoke about my DCSE internship, where I was the team lead of a module. Then I was asked about my performance as the lead and how I helped my team develop our module. She also asked me about the things I did as the lead that I wouldn't have typically done as a team member.

Then I was asked to talk about my most favourite project from the resume. I spoke about my Full stack project (neighbourhood service finder). I was asked about how it stood out from similar platforms like urban and also the struggles I faced while developing it.

Then I was asked to solve two puzzles.

1. Why are manholes circular in shape and not a square or a rectangle?

   I answered this by saying how a circle has no corners and hence it could be the shape that could get the least damaged. I also explained that it could ensure faster flow of water, than a shape with an edge or a corner.

2. Cut a circular cake into 8 pieces (not necessarily equal) using only 3 straight lines.

   (She offered me a paper and pen to work out, in case of need)

   I tried different ways but I was not able to answer this. So, I told her and then she moved on to the next question.

Coding questions:

I was asked two coding questions (easy-level).

1. Finding odd numbers from a stream of numbers.

   I was asked to write the code for it. I explained the code as I wrote. I used modulo operator to identify the odd numbers and stored them in an Array list instead of an array, which has fixed memory (as we do not know the total number of elements).

   CODE:
   ```
   List<Integer> findOddNumbers(List<Integer> stream) {
       List<Integer> oddNumbers = new ArrayList<>();
       for (int num : stream) {
           if (num % 2 != 0) {
               oddNumbers.add(num);
           }
       }
       return oddNumbers;
     }
   ```

2. Then I was asked to solve the two sum problem (with a slight variation).

   Given two arrays, generate and return pairs of elements such that the first element is from the first array and the second element from the second. The catch was that the sum of these two numbers should be equal to a given target value.

I asked if I could write the logic in pseudocode, she said yes.

I started with the bruteforce solution, where I scanned each element in the first array and computed its sum with each element of the second array. If they added up to the target, I stored them in a 2D array (I had a counter to store elements in the 2D array) and then finally returned the 2D array.
The complexity of this would be **$O(n^2)$** (product of the sizes of the two arrays).

To optimize it,
I used a hashmap to store the elements of the 2nd array.
I scanned each element of the first array and I computed the complement by subtracting the element from the target sum, and then I checked if the key of this complement was present in the hashmap.

If it was present, I added them to a 2D array and finally I returned the 2D array.
The complexity of this would be **O(n)** (size of one of the arrays).

Then I explained the bruteforce approach. And when I started to explain how I optimized it, she said that bruteforce was enough but was curious to see how I optimized it.

So I explained how I optimized it as well, she liked my answer.

CODE:

Bruteforce approach:

```
List<int[]> findPairsBruteForce(int[] arr1, int[] arr2, int target) {

List<int[]> result = new ArrayList<>();

for (int i = 0; i < arr1.length; i++) {

for (int j = 0; j < arr2.length; j++) {

if (arr1[i] + arr2[j] == target) {

        result.add(new int[]{arr1[i], arr2[j]});

}

}

}

return result;

}
```

Using hashset:

```
List<int[]> findPairsOptimized(int[] arr1, int[] arr2, int target) {
    List<int[]> result = new ArrayList<>();
    Set<Integer> set = new HashSet<>();

    for (int num : arr2) {
        set.add(num);
    }

    for (int num : arr1) {
        int complement = target - num;
        if (set.contains(complement)) {
            result.add(new int[]{num, complement});
        }
    }

    return result;
}
```

Role based questions:

1. I was asked to write test cases for a date of birth field, which had to validate the age of the user to be between 18 and 65. I answered this confidently as I had prepared for similar type of questions. I covered all possible cases.
2. Next question was 'If a server failed only on Fridays and worked well on all other days, what could be the possible reason?'. I explained about the possibility of an error in some logic involving the 'day' field and hence it only failed on that particular day.

Finally, I was asked if I had any questions for them.

I asked them about one of the projects mentioned in the PPT, involving document handling. I asked about this, as I had a project idea similar to it. She pulled up info related to it on her laptop and gave me a glimpse on the project. It sounded interesting.

This round focused on aptitude, coding and role-based questions.

Technical Interview – 2: (10-20 mins)

The interview was conducted on Google Meet, and the interviewer joined virtually.

He asked me to introduce myself first.

Then I was asked about why I had chosen the role of QE, instead of SE. This question seemed to be important, as it was also asked in the HR round.

He asked the difference between a tester and a quality engineer. I told them that a tester just reports errors, but a QE is also focused on improving the quality of the software right from the initial stages of development.

Then, I was asked about the types of testing (functional and non-functional). I also mentioned black box testing, white box testing and grey box testing. I was asked to explain each of the 3.

He then asked me why I chose Appian. I explained how I liked the idea of automating work without heavy coding and how the idea of minimal code motivated me to apply (Research about the company before the interview).

Next question was a bit tricky, he asked me if I would choose SE if I had the option. I said no, because it's usually advised to stick to the role you have been shortlisted for (as you might get rejected directly and also if you aren't shortlisted for any other roles, you may lose your opportunity.)

Finally, he asked me if I had any questions for him.

Since my role would involve a lot of testing and it was mentioned in the PPT that they use python tech stack for testing, I asked about the tools or concepts in python I would have to focus on learning, as I had only basic knowledge in python. He said that we would be given full training in the beginning of the internship and hence I wouldn't have to worry about it.

This round focused on both technical and behavioural aspects.

HR round (5 – 10 mins):

I was first asked to introduce myself quickly. Then some background questions and about my family.

I was also asked about my hobbies, which were mentioned in my resume. He asked why I chose this role and again, why I preferred this to SE.

He asked if I was in the top 10 or top 20 of my batch, based on my CGPA and then he asked if I had any questions for him.

The result were announced around 9pm the same day, and I was one of the 3 candidates who got the offer for QE.

General Tips:

Stay calm through out the process. Don't let your anxiety take over you.

Have a neutral mindset, don't let anything get to your head.

Peer pressure and insecurities are challenges that you can get through only by believing in yourself.

Answer all questions honestly and confidently.

If you are unsure of something, admit it. They are looking for candidates, who are willing to learn.

Practice DSA fundamentals (arrays, strings, stacks, linked lists, queue, tree), OOPs and DBMS concepts for coding assessments.

Understand these concepts and try to apply them in real world scenarios.

Practicing puzzles and general aptitude questions will build your critical thinking skills.

Be thorough in all the projects you mention in your resume.

If you need a second to think, you can very well say that you need some time to think and then answer.

The way you think and approach matters more than the solution you compute.

Be clam and talk slowly during the interview.

Wear professional attire and look neat.

Here's hoping you get an internship too, and get to write your own story like this.

Roopa Varshni R,

CEG CSE'27

Wishing you all the very best :)

Never stop believing in yourself. It'll be worth it (●'◡'●)

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